Rank-27 BVE Flat-Slice Representation Repair

Tactix Development Project

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Executive summary

This increment removes per-clause heap ownership from BVE candidate materialization without changing BVE's proof rule or scheduling. Positive and negative antecedents, generated resolvents, and their LRAT chains now occupy reusable append-only flat pools. Small records retain offset/length slices, not pointers, so pool growth cannot invalidate them. The production diff has SHA-256 `63ea569e4d504964b14bd49a647a6bba13b187367cd28a1da4b171f664cdadef`; the validated Release binary has SHA-256 `aeb14e2a075c58db313187df98de807fb9c9a3a5abccb8fbaab57eb42a677f62`.

Every exact-stage image preserves antecedent, candidate, pair, literal, proof-chain, tick, cost, bound, scheduling, learning, pruning, and retained-state order. No benchmark identity, rank, deadline, per-instance arm, or tuned constant appears in the solver. T105 cuts BVE time 108.667 to 52.943 seconds; the complete frozen B28+C5 gate passes 66/66 at 0.983720x predecessor PAR-2; and the serial Rank-27 control cuts BVE 37.975 to 16.229 seconds and total time 498.082 to 419.772 seconds. Three same-node controls measure 371.711, 375.797, and 372.969 seconds, versus the 386.665-second promoted-Vivify baseline. Their median is 372.969 seconds; BVE averages 15.481 seconds with only a 0.030-second range. The 300-second milestone remains open and will not stop further portable convergence work.

Claude Code's authoritative terminal report classifies exact-source fuzz job `1647571` PASS: 27/27 corpus entries and 10,000/10,000 random cases have zero failures, rejected models or proofs, differential disagreements, sanitizer findings, crashes, aborts, or invariant failures. All required BVE, BveEliminated, and Propagate reach rows fired, and certificate/scratch residue is zero.

1 Representation and boundary preservation

The prior implementation stored every materialized antecedent and every generated resolvent in its own `std::vector`. Rank 27 performs 1.699 million BVE applications and 64.243 million resolution attempts on the accepted trajectory, making allocator ownership an avoidable part of the hot path. Kissat instead uses raw reusable stacks for the same lifetime.

Tactix now keeps four reusable payload pools in the existing per-rule `BveDefScratch`: antecedent literals, absorbed root-unit proof IDs, resolvent literals, and resolvent LRAT chains. Antecedent and resolvent records carry a pool pointer plus 32-bit offset and length. Offsets remain valid across vector relocation. Pools are cleared only when the next candidate begins, after the prior candidate has committed or declined. A satisfied large antecedent rewinds both captured pool starts; tautological and oversized resolvents rewind their literal start before the old control flow continues.

This is an allocation representation change, not a new rule. The existing antecedent admission, gate classification, pair traversal, resolvent-size and growth bounds, proof transaction, elimination reconstruction, and policy schedule are untouched. Every registered BVE entry already owns the persistent scratch; a debug assertion makes that lifetime requirement explicit. The public N1 operation

remains complete, N2 retains independently checked models and proofs, and neither N3 nor N4 is weakened.

2 Exact mechanism and correctness gates

The first full-length mechanism witness is T105 job `1647289`. Removing timing lines makes predecessor and candidate byte-identical at SHA-256 `7f9530cc51c77fcee0bf93ce108b540c997bc1338fc9cc49d580f626f99ebd34`. That image includes 461,267 conflicts, 6,178,985 BVE applications, every resolution/effect/tick/bound counter, the complete rule schedule, learning, pruning, and the terminal UNSAT verdict. The measured change is:

T105 measure	Predecessor	Candidate	Delta
Internal solve, s	426.944	364.015	−62.929
BVE, s	108.667	52.943	−55.724
Inprocessing, s	286.971	228.413	−58.558
Peak RSS, KiB	3,079,004	3,078,884	−120

Frozen full-width campaign `1647325/1647380` executes the complete Tier B plus predeclared mechanism-relevant C5 under two CPUs, 16 GiB, and the 1,200-second solver contract. All 66 tasks are valid. Each arm independently checks 17 SAT models and 16 UNSAT proofs against the original decompressed CNF; all 33 paired normalized non-time images match. Candidate PAR-2 is 1514.815 seconds versus 1539.885, ratio 0.9837196. Invalidity, solve loss, cleanup failure, and certificate residue are zero. Summary SHA-256 is `338b66b1fb7c66bbfdccfc7e1a3cdac5902475b51adddc47b4dbefd54ee6549a`.

Proof-producing Rank-27 job `1647572` follows the same exact trajectory. Pinned `drat-trim` verifies the 1,514,395,111-byte proof in 146.366 seconds with `s VERIFIED`. Proof SHA-256 is `0f99f55ac55571d6f83fe53ec8cb4a6a4c8df0a6e127442e50f47e25bea8bb2d`; the normalized proof-mode image is byte-identical to the accepted predecessor. The proof, decompressed input, checker output, and scratch are deleted after verification, leaving zero residue.

3 Rank-27 performance and disposition

Serial exact-state campaign `1647569` compares the promoted-Vivify pin and the BVE candidate in one allocation. Both complete non-time outputs have SHA-256 `14f7da2d94153a2afd1776ecab8e7e49e5c2c0613dc26c43714471b7ecbe6194`. Thus all 346,693 eliminations, 64,242,959 resolutions, definition/gate/tick and bound counters, rule schedule, search, learning, pruning, and retained state are exact. BVE falls 37.975 to 16.229 seconds (0.42736x), inprocessing falls 182.996 to 150.772 seconds, and internal solve falls 498.082 to 419.772 seconds. The 5,564-KiB RSS increase is 0.88% and remains far below the 16-GiB contract; it is the bounded retained pool capacity. Cross-node outside-rule time is not assigned causally.

The user-directed same-node control `1647573` provides the cleanest causal comparison. The BVE candidate reaches exact-state UNSAT in 371.711 seconds with 15.470 seconds in BVE, versus promoted-Vivify job `1647466` at 386.665 and 30.069 seconds. Total improves 14.954 seconds and BVE improves 14.599 seconds; outside-inprocessing time is flat. Exact-binary repeat `1647575` completes in 375.797 seconds with 15.500 seconds in BVE. Its 4.086-second variation from the best run contains only 0.030 seconds of BVE variation and independently confirms the owner-specific gain. All staged inputs and certificate/checker scratch are absent. Third exact-pin job `1647654` lies between them at 372.969 seconds total and 15.472 seconds in BVE. It is 1.258 seconds slower than the best, but only

0.002 seconds of that delta is in BVE; 1.097 seconds lies outside inprocessing. Across all three runs the whole-run range is 4.086 seconds and the BVE range is 0.030 seconds, so job `1647573` remains the single best while the causal BVE gain is repeatable.

The clean Release suite passes 73/74: every executable, solver, proof/model, trace, N1, proof-rule/strategy-boundary, campaign-cleanup, and solve-reset test passes. The sole failure is the pre-existing opt-in documentation-size policy; the repository authority forbids automatic compaction. The asynchronous full Kissat Tier-D campaign remains non-scorable until its aggregate classifies all invalidity and solve loss; this increment neither waits for it nor interprets partial resource exits as solver failures.

The authoritative fuzz PASS completes the BVE promotion evidence. The already-frozen active-decision representation is the next portable search increment: it preserves the exact Rank-27 image, completes in 373.696 seconds, and passes its own full B28+C5 guard 66/66 at 0.995586x PAR-2. It remains separate from this commit until the BVE checkpoint lands. Beyond that increment, freeze the newly accepted baseline and current-Kissat comparator, then localize collection/rewatching by substages and normalize by entries/literals processed before evaluating the bounded fused compact-and-forward traversal. Next localize matched Substitute invocations, then revisit the smaller residual Congruence difference and continue broad search-quality convergence. Each remains a separate prefix–Rank-27/B+C–full gate–Claude fuzz–report increment. Do not reopen the closed Sweep, counted directed bridge, BVE antecedent-admission, accepted Congruence, or accepted Vivify repairs without contradictory exact-state evidence.